

Flight Phase: **Climb**

Input name	Value	Label	Degree
Turbulence	1.84	Moderate	0.73
Air speed	222.53	High	0.25
Altitude	5964.57	Medium	0.10
Pitch angle	-8.00	Nose down	0.30
Vertical speed	21.42	Level	0.98
Engine health	1.80	Critical	0.80
Fuel status	89.61	Full	0.97
Proximity to obstacles	476.96	Caution	0.71
TCA's conflict level	0.54	Advisory	0.31

Output name	Value	Label	Degree
Thrust command	77.50	High	0.83
Elevator deflection	-15.00	Down	1.00
Autopilot mode suggestion	0.92	Alt hold	0.96
Stall warning	1.00	Advisory	1.00
Fuel reserve command	1.00	Economy mode	1.00
Stress reduction command	0.92	Increase autopilot	0.97

Explanation
- Low airspeed and pitch attitude indicate high stall risk; issuing stall advisory is critical. - Vertical dynamics suggest aggressive altitude change, requiring corresponding thrust and control.

Main Contributing Factors		
Vertical speed	21.42	Level
Fuel status	89.61	Full

Flight Phase: **Landing**

Input name	Value	Label	Degree
Turbulence	2.80	Severe	0.61
Air speed	114.24	Slow	0.79
Altitude	488.12	Low	1.00
Pitch angle	15.21	Nose up	0.96
Vertical speed	1912.87	Climb	0.59
Engine health	0.69	Degraded	0.59
Fuel status	5.22	Empty	0.96
Proximity to obstacles	76.63	Immediate danger	0.47
TCA's conflict level	2.23	Resolution advisory	0.75

Output name	Value	Label	Degree
Thrust command	100.00	Max	
Elevator deflection	-20.00	Down	
Autopilot mode suggestion	0.00	Manual	
Stall warning	1.00	Warning	
Fuel reserve command	0.00	Emergency diversion	
Stress reduction command	1.00	Override manual	

Explanation
- Low airspeed and pitch attitude indicate high stall risk; issuing stall advisory is critical. - Engine health degraded; thrust adjusted to ensure performance under failure conditions.

Main Contributing Factors		
Altitude	488.12	Low
Pitch angle	15.21	Nose up
Fuel status	5.22	Empty

FAIL SAFE MODE ACTIVATED

A critical condition was detected
Emergency control outputs were issued.
Fuzzy logic system was overridden for safety response.